ISOMETRIC

Biomass or Bio-oil Storage in Salt Caverns

Storage Module

v1.1

This module details durability and monitoring requirements for bio-oil or biomass storage in salt caverns.

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1.0

Introduction

Durability refers to the length of time for which CO_2 is removed from the Earth's atmosphere and therefore cannot contribute to further climate change. This Module details durability and monitoring requirements for bio-oil and biomass storage in salt caverns.

Salt caverns have historically been used to store or dispose of a wide variety of materials including hydrocarbons, brine, industrial and even nuclear waste. Salt's viscoelastic properties make caverns nearly impermeable to emplaced materials, reducing the risk of migration. Additionally, under stress salt deforms slowly resulting in the cavern closing around the wastes over long time periods and specific cavern conditions, entombing them within the low permeability salt[1]. The geometry of salt caverns influences their storage capacity, stability, and integrity. Homogeneous salt caverns are within salt domes or thick salt beds and are solely surrounded by salt. Their stability and integrity are based on the rock salt only. Inhomogeneous salt caverns are within bedded salts or salt breccias, their stability and integrity are based on the properties of rock salt and non-salt permeable structures.

This Module is applicable for bio-oil or biomass slurry emplacement into salt caverns that have been approved by the relevant permitting authority. Storage within salt caverns can occur when the salt cavern is no longer viable for high pressure natural gas cycling or compressed air storage, or other waste storage. The emplacement and storage of fluids and waste products into salt caverns has been occurring since the 1940s and 1950s respectively[2].

Bio-oil is a dark, viscous liquid typically between pH 2-3 (but up to 6), consisting of oxygenated hydrocarbon compounds[3]. Prior to emplacement, the pH may be buffered and/or salinity raised as required by permitting. Bio-oil can have co-products like biochar mixed into it ahead of injection underground. Within this Module, we use the words 'bio-oil', 'bio-oil with biochar' and 'injectant' interchangeably. The storage of bio-oil in salt caverns is relatively new and has not been well studied and documented as of December 2023.

Biomass slurry is an organic waste (e.g., manure, food waste, agricultural waste, paper sludge) mixed with on-site brine. The slurry contains compounds like carbon, nitrogen, phosphorus, oxygen, hydrogen, sulfur, and trace elements found in the organic waste.

As both bio-oil and biomass (hereafter known as "the injectate") are expected to be denser than the surrounding subsurface brine in the cavern it is expected that they will sink to the bottom during emplacement. Displaced brine is pumped out of the cavern and injected back into a different subsurface reservoir. The durability of the injectate stored within salt caverns depends on the characteristics of the injectate, the salt and any bedded layers, and the interactions between the two being well defined and monitored. This, when coupled with capping and closure of subsurface reservoirs as per the U.S. EPA Underground Injection Control (UIC) or equivalent permitting requirements, removes the CO_2 stored within the injectate from the atmosphere for geological timescales[4] [5] [6] [7].

Potential risks to the expected durability of biomass and bio-oil are as follows:

- Emplacement may cause cavern wall dissolution.

    - Any liquid within the emplaced material that is below salt saturation may cause dissolution of the cavern walls. Prior to cavern emplacement, material must be salt saturated as required by the governing UIC or equivalent permit. For example, salt crystals may be added commensurate to bio-oil water content and biomass may be mixed with displaced cavern brine to minimize salinity differences within the cavern and mitigate dissolution.
    - The low pH of bio-oil can induce reactions with the cavern infrastructure, which may decrease durability by providing conduits for migration out of the cavern. Mineral blankets can be used to protect the tubing string from continuous exposure to emplaced materials and/or the bio-oil buffered to a higher pH.
- Injectate may be converted to bio-gases within the cavern such as CO_2, CH_4, N, O_2, H_2S and VOCs.
    - The emplaced material could biodegrade to form CO_2, CH_4, N, O_2 and VOCs and other short-chain hydrocarbons. Microbial activity could also cause H_2S formation. Gas formation may increase the pressure within the cavern decreasing the stability. Any signs of gas formation should be monitored as part of the operation and post-emplacement monitoring plan (see Sections 3.1.3 & 3.2). Any gases produced will be considered removed because of the impermeable cavern walls and roof and well design preventing the migration of buoyant phases (such as biogas). Any releases of C bearing gasses from the salt cavern must be deducted from the net CO_2e removal (see Section 3.3).
- Emplacement may lead to a loss of cavern integrity, for example via pressure and/or temperature differentials

which could lead to fluid expansion/reduction and fracturing on closure.

- Differences in cavern pressure compared to the surrounding formation could result in the deformation of salt, for example via fracturing, salt creep or roof collapse. This could result in subsidence, the leakage of the injectate outside of the cavern and to USDWs/the atmosphere and/or a decrease in cavern volume. As the injectate will be denser than the brine it is replacing it is likely to increase the cavern stability. Any signs of pressure or temperature differentials must be monitored and emplacement/closure procedures adapted accordingly. The overlying strata can also provide support for the overburden reducing the risk of roof collapse.

This section outlines requirements for evaluating emplacement and storage, with a focus on cavern characterization, infrastructure construction and monitoring. The post-emplacement monitoring plan detailed in Section 3.2 acts to address and mitigate these potential risks to durability. Section 3.3 addresses accounting for any GHG emissions associated with monitoring the geologic storage of CO_2 during the project operations, closure, and post closure periods.

Monitoring of the emplacement cavern, and overlying formations and surface where applicable, must be completed to ensure that any emplaced material remains stored within the confines of the salt cavern and does not migrate outside of the cavern, nor result in decay of the injectate and subsequent re-emission as CH_4, CO_2 or other volatiles. The emplacement cavern, and overlying formations and surface where applicable, must be monitored in accordance with the U.S. EPA Underground Injection Control (UIC) or equivalent permitting requirements as specified in the operating permit

for the salt cavern issued by the relevant regulatory body. This Module addresses minimum requirements for emplacement well design, construction, operation, and monitoring to ensure proper cavern storage design construction and monitoring to ensure durability of storage. Although typically these requirements will be addressed in full in the UIC permit or equivalent, the emplacement of bio-oil and biomass into caverns is a novel approach for which permitting decisions and requirements are still in development and may not be consistent based on the evaluation and development of permitting approaches by each responsible authority. Therefore, critical concepts and requirements are documented here for consistency and to ensure proper cavern storage design construction and monitoring to ensure durability of storage.

The subsurface monitoring approach developed and implemented by the Project Proponent or Operator (when the Project Proponent is not operating the cavern) must address, via the permitting process and permit compliance, or by additional efforts and documentation:

- Salt Cavern Characterization: the proposed storage cavern, surrounding areas that could or would be affected by the salt cavern's integrity, overlying formations and Area of Review[8] (AOR) must have been adequately characterized to demonstrate cavern suitability for storage and containment of the injectate. If an AOR is not defined within the permit, the Project Proponent/Operator must define an AOR. See Section 2.2 for further details.
- Emplacement Infrastructure Construction and Performance: the proposed storage cavern infrastructure and emplacement system must be designed in accordance to the UIC or equivalent permit and relevant API and ISO standards (e.g., API standard 65- part 2 on

"Isolating Potential Flow Zones During Well Construction"[9]), including design and specification of wellbore and well materials to ensure proper long term operation of the well when injecting and protection of the lowermost underground source of drinking water (USDW).

- Emplacement System Operation & Monitoring: the Operator must specify operating conditions and monitoring systems and approaches, such as gas detection, and other systems to ensure that the emplaced material remains in the cavern, the salt cavern is not negatively impacted by operations, automatic safety precautions are in place to minimize potential for exceeding allowable operating conditions, and conditions can be monitored for compliance or deviation from requirements. Reporting of operations will be in accordance with the UIC or equivalent governing body.
- Closure and Post-Closure Requirements: Requirements for proper closure of the salt cavern and emplacement facility, as well as post closure requirements and post-emplacement monitoring to ensure the injectate remains sequestered durably in the cavern and poses no net environmental harm, the cavern is properly monitored, and any non-compliance is addressed with corrective actions.

Specifically, the following requirements must be met to ensure durable storage of injectate in the salt cavern.

2.0

Permitting and Cavern Characterization

2.1

Permitting

The emplacement cavern must have a current Class V UIC well permit or equivalent issued by the responsible authority for the location of the emplacement facility and salt cavern. The permit must specifically identify biomass, bio-oil or an equivalent type of injectate, as acceptable injectates under the permit[9].

2.2

Cavern Characterization and Feasibility Requirements

The cavern should be characterized in accordance with the permit application and approval requirements under the UIC or equivalent regulations. Cavern characterizations must include evaluation of cavern chemistry and conditions, where required, to ensure compatibility of the injectate with the salt cavern.

As part of the permit application, the Operator must demonstrate, where applicable, that the cavern:

- Is of sufficient volume to receive the total anticipated volume of the injectate stream;
- Is free of transmissive faults and fractures and of sufficient extent and thickness to contain the emplacement volume, displaced cavern brines and any biogas generation;

- Has structural integrity (i.e., cavern walls are of an adequate thickness and extent and it is not in close proximity to other caverns or the salt boundary, no evidence of salt creep, salt has sufficient compressive and shear strength) such that it will prevent migration of the injectate, brine or biogas above the storage complex, towards the surface and atmosphere and/or USDWs;
- Low permeability and thickness of cavern to prevent vertical or lateral migration of the injectate, brine or biogas above the storage complex, towards the surface and atmosphere and/or USDWs;
- The operator must also identify a suitable formation for displaced brine injection.

In addition, the Project Proponent must also characterize the following to assess the risk of leakage, develop the operation conditions for injection and monitoring plans, model the injectate behavior and for comparison to future measurements:

PARAMETER	PURPOSE
Volume and geometry of the salt cavern	To demonstrate the capacity of the cavern to receive and safely store emplaced material, and to help determine operating pressures.
Confirmation of minimum cavern wall and roof thickness and distance from edge of salt or other caverns as set by the regulating body	To demonstrate cavern integrity.
Confirmation of low permeability, lack of highly soluble minerals and structural integrity of salt cavern	To demonstrate lack of migration pathways and that any emplaced material will be trapped and unable to migrate out of the cavern
Identification and characterization of porosity, permeability and mineralogy of the salt cavern "host" rock, and any interbeds of non-salt layers that may be present	To demonstrate lack of migration pathways.
Cavern specific geotechnical characterization to ensure cavern integrity. For example, things that would typically be done during pre-construction like determination of rock mechanics and regional stress, including strength measurements (e.g.,	To determine Cavern integrity and potential risks.

PARAMETER	PURPOSE
unconfined & confined compressive strength, constant mean stress compression and extension tests, tensile strength) of the cavern walls and overlying formations, compression tests, dissolution tests, overburden/horizontal stress, constant stress creep tests.	
Temperature, pH and conductivity/chloride concentration of the cavern brine.	To determine optimum conditions for cavern stability and identify potential interaction of the injectate under these conditions with the storage complex which may impact whether any potential products (e.g., biogas) are produced and injectate stability.
Total carbon © content in the cavern brine	To determine baseline concentrations that can be compared to during operation to identify any carbon removal on brine pumping.
$\delta^{13}C$ of the compounds of the injectate, where applicable	For determining the source of any produced biogas and extent of reactions (e.g. methanogenesis) as a result of injection.
An assessment of the potential for ground subsidence and if necessary an establishment of a pre-emplacement baseline of topography or reference points for subsidence over the Influence Area[10], including locations of data collection points, and current subsidence rate values.	To determine the risk of ground subsidence and determine a baseline topography across the area of Influence.

Salt cavern characterization can be provided in the form of previous characterizations and historical data, as long as the requirements outlined within this Module are met.

The Project Proponent or Operator (when the Project Proponent does not operate the cavern) is required to demonstrate that there is an approved permit for displaced brine injection in place. It is a requirement that re-injection of the brine must occur within a closed system to avoid any contact and equilibration with the atmosphere, resulting in potential reversals. The Project Proponent is required to assess and quantify any potential reversals as a direct result of brine injection.

Cavern characterization is required to be reviewed every five years as part of the project Crediting Period (see Isometric

Standard) at a minimum, or at the UIC Programs Director's or equivalents request, or when monitoring and operational conditions warrant, as indicated by a significant change in site conditions, injectate characteristics, or monitoring data. The review must include a comparison of pre-emplacement project assumptions to actual measured conditions including salt cavern capacity, the structural integrity of the cavern, and specific operating conditions observed during emplacement. Estimates revised with any acquired monitoring data should demonstrate that the planned emplacement volume will remain within the salt cavern until the end of the post-emplacement monitoring period.

2.3 Well Construction Requirements

The Project Proponent or Operator (when the Project Proponent does not operate the cavern) must ensure that the emplacement well is constructed in compliance with the EPA UIC or equivalent permit. All documentation and records of well construction are required to be maintained and available for review, at any time during the project lifetime.

At a minimum, the Operator must ensure that any legacy wells which may exist within the delineated AOR have been evaluated and wells which pose a risk to durability are properly plugged prior to emplacement in order to:

- Prevent the movement of fluids into or between any unauthorized zones
- Prevent the movement of fluid into USDW
- Permit the use of appropriate testing devices and workover tools

- Permit continuous monitoring of the emplacement well pressure in the annulus space between the emplacement tubing and long string casing

Casing, cement, tubing, packer, wellhead, valves, piping, or other materials used in the construction of each well associated with the project must have sufficient structural strength and be designed for beyond the life of the project. All surface casing will be set below the lowermost USDW and cemented to the surface. All well materials must be compatible with fluids with which the materials may be expected to come into contact, including the injectate and cavern brines (e.g., corrosion-resistant well casings) and must meet or exceed standards developed for such materials by API, ASTM, ISO, or comparable standards. Utilized standards are required to be clearly outlined within the project design document (PDD) submitted by the Project Proponent. The casing and cementing program must be designed to prevent the movement of fluids out of the sequestration zone and above the storage complex.

If pre-existing wells are being used for injection or monitoring, special considerations are necessary to ensure the integrity of the well and to prevent fluid migration along the borehole. These must be agreed and checked by the regulating body to determine construction and safety is consistent with well construction requirements. All checks and modifications must be recorded and all records kept. This must include the following:

- determining the integrity of the cement and casing.
- conducting any necessary action to repair defects.
- determining whether the existing well materials are adequate for the new function of the well.

- determining the diameter of the hole, any deviations from vertical, and any significant curvature or bends in the well should be compared with the size of the proposed monitoring equipment.
- existing well materials should be checked to ensure that they are compatible with fluids with which they will come into contact, such as carbon dioxide and carbon dioxide-rich brines if they are completed in the injection zone.
- any flaws in the casing or cement will need to be repaired.

3.0

Monitoring

3.1

Emplacement and Operational Monitoring Requirements

The Project Proponent or Operator (when the Project Proponent does not operate the cavern) is required to ensure that the emplacement facility complies with the well permit, including the development and implementation of the well operating plan as required by the permit. The Project Proponent must monitor the composition of the injectate and ensure it complies with the relevant permits. All other monitoring is required by the cavern Operator (or Project Proponent if they are operating the cavern) to ensure all material emplaced into or produced from the cavern must be sampled and analyzed in accordance with the approved written waste analysis plan required by **the authorizing agency**. At a minimum, the permit and associated well operating plan must consider the following:

3.1.1

Injectate and Operational Monitoring

- Emplaced material must be gravity fed or emplaced at a surface pressure beneath that which is specified in the permit, ensuring the cavern does not become over-pressurized.
- The volume of emplaced material must be monitored appropriately. For example by monitoring the rate of injection or by using the weight of biomass/bio-oil

emplaced and as well as the amount of water added where applicable.

- Limitations on composition of the injectate, including, but not limited to pH, temperature, conductivity/chloride concentration, if relevant, to ensure the injectate does not negatively impact the emplacement well, associated facilities and the cavern walls via inducing dissolution, reaction, or other degradation pathways, resulting in increased potential for migration of the emplaced material.
    - Records of laboratory analyses and relevant permit limitations to demonstrate compliance must be maintained in accordance with the well permit and available for review at any point during the Crediting Period or post closure.
- Analysis of the injectate with sufficient frequency to yield data representative of its chemical and physical characteristics, sufficient to demonstrate there is minimal effect on the cavern. In addition, analysis of the injectate must demonstrate compliance with the well permit and be available for review. Injectate analysis should consist of the following parameters using industry standard or indicated methods using properly calibrated equipment:
    - Total C content (for further details see Section 7.3.3.1 of the relevant Protocol)
    - pH
    - Temperature
    - Chloride concentration or alternative determination as required by the regulator
    - Density (bio-oil)
    - Total acid number (TAN) (bio-oil)
    - Water content (bio-oil)

For all injectate monitoring and analyses, sufficient samples must be analyzed to determine that the composition of the injectate is within specified parameters in the UIC permit or equivalent, where required.

For samples taken each emplacement batch, each individual batch that is emplaced should be analyzed and characterized to ensure composition variation from batch to batch is accounted for. Samples should be well mixed and representative.

For samples measured per feedstock type, a representative value should be used. These measurements should be repeated to find representative values every time there is a material upstream process change like a new biomass feedstock. If a blended feedstock is emplaced, samples should be taken for each emplacement batch.

Wells must have gas detectors (or equivalent sensors/imaging) with alarms and injection shut-off systems (e.g., automatic shut-off or procedures in place for manual shut off of injection/operation), including for a gaseous release (CO_2, hydrocarbons, or other GHGs) and injection pump shutoff when maximum pressure is reached or maximum flow rate is exceeded. If the injectate is not in a gaseous phase, then detectors/alarms may be placed on any producing wells (e.g., brine producing wells/tanks for salt caverns) as an alternative to wellhead monitoring. Wellhead monitoring is then required if gas is detected and is found to be a result of biogas formation. If gas detection alarms are activated, the operator must immediately investigate and identify as expeditiously as possible (or in accordance with permit requirements) the cause of the alarm or shutoff, and report the instance to the validation and verification body (VVB).

3.1.2

System Integrity Monitoring

- Corrosion monitoring of well materials (internal mechanical integrity), upon well completion and performed and reported at a cadence set by the UIC permit or equivalent permit thereafter, for loss of mass, thickness, cracking, pitting, and other signs of corrosion, to ensure that well components meet the minimum standards for material strength and performance set by API, ASTM International, or equivalent. If emplacement frequency is less than twice per week in that timeframe, then corrosion monitoring requirements are not mandatory, but are recommended. This could, for example, be done by measuring the pH of the cavern fluids to ensure it is not corroding the wellbore, through corrosion surveys in the production tubing or through laboratory experiments exploring corrosivity[11].
- A demonstration of external mechanical integrity on a cadence specified by the UIC permit or equivalent until the emplacement well is plugged and prior to abandonment.

3.1.3

Migration and Storage Reversal Monitoring

- Continuous or high frequency monitoring of cavern pressure to ensure the cavern does not become overpressurized thus reducing the risks to the cavern integrity and this leakage.
- Periodic sonar surveys should also be conducted to quantify the emplaced material and confirm containment as well as any salt creep that may be occurring (cavern

geometry). At a minimum this should be done prior to emplacement, and when cavern capacity is expected to be at set points based on cavern specific data, for example at 33% and 66% full or at 50% full, and in agreement with regulators. The Operator may use an alternative but equivalent method than sonar surveys if agreed with the Project Proponent (if different to the Operator) and with the regulating body. This must be documented in the PDD.

- Cavern fill monitoring is required to provide information about the sump depth and its evolution, for example by comparison of the volume emplaced and cavern volume, that is confirmed quarterly by a depth to top of emplaced volume check.
- Analysis of the displaced brine stream with sufficient frequency to yield data representative of its chemical and physical characteristics using industry standard or indicated methods and quality and properly calibrated equipment:
  - Volume (continuous e.g., by flow rate)
  - pH
  - Conductivity/chloride concentration
  - Total organic carbon concentration
- Monitoring of the composition of any gas recovered in the displaced brine (or monitoring wells or representative sampling locations when available) is required where applicable, when there is a detection of gases from the cavern. Gas monitoring must include CO_2, CH_4 and VOCs emissions from the salt cavern via gas monitors with a resolution of at least 0.01 vol%. When concentrations above background atmospheric levels are detected, a sample must be taken to establish the chemical composition of the displaced gases (including CO_2, CH_4, N_2, O_2 and VOCs).

    - Results must be compared to baseline values obtained prior to emplacement, differences must be assessed by the Project Proponent to see if they are attributable and material to the project. There are two options for determining a baseline, once the liquid petroleum gas (LPG) has been vented and gas measurements plateau following a spike from venting:
        - Once the LPG has been vented, if the gas measurements plateau and no gas is detected, the background will be zero and any gas later detected will be attributed to the project.
        - Once the LPG has been vented, if there is still detectable gas during the gas measurement plateau, the baseline should be taken as the plateau measurement following LPG venting.
    - If no baseline was established after venting, any gas later detected will be attributed to the project.
- Surface subsidence monitoring (every 2 years), for example using GPS or a set of benchmark levels on the storage cavern, is required and will be compared to baseline data and trends. This will be used to monitor for indication of salt movement (and volume variation) or cavern roof collapse. The surface elevation change of concern must be agreed with the regulating body prior to the start of emplacement.
- A seismic monitoring program may be required if determined necessary within the permit, to demonstrate that seismicity will not impact the cavern and durability of the injectate. The area within the AOR of the injection facility and the area of the storage complex must be monitored for magnitude 2.7[12] or greater seismic events to determine the presence or absence of any induced micro-seismic activity associated with the wells and near any discontinuities, faults, or fractures in the subsurface.

For all emplacement monitoring and analyses, sufficient samples must be analyzed to determine that the composition of the injectate is within specified parameters in the UIC or equivalent permit, where required. Each individual batch of injectate that is emplaced should be analyzed and characterized to ensure composition variation from batch to batch is accounted for. Samples should be from a well mixed and representative container of the injectate. Requirements for C content analysis are set out in Section 7.3.3 of the relevant Protocols.

If any leaks are detected from the cavern, the Project Proponent/Operators must undertake corrective measures as set out in their monitoring plan submitted and approved by the permitting authority. If the cavern is found to have lost integrity, the Operator must halt emplacement whilst they conduct an assessment to determine whether there are any leaks and whether the loss of containment and/or well mechanical integrity and/or cavern integrity can be repaired prior to recommencing operations. The amount of CO_2 lost must also be quantified and subtracted from the overall total of CO_2 stored.

Re-evaluations of the emplaced material must also be implemented when warranted based on observational or quantitative changes of the monitoring parameters of the salt cavern, including but not limited to:

- Loss of cavern or well integrity and migration of the injectate outside the salt cavern
- Materially elevated pressure/temperature

3.2

Post-emplacement Monitoring Plan

The aim of this post-emplacement monitoring plan is to put in place scientific and/or operational monitoring practices which go beyond requirements of the UIC or equivalent injection permits specified under the UIC or equivalents well classes allowed under this Protocol. This is in order to prove beyond reasonable doubt that storage is expected to be durable on geologic timescales at which point the cavern can be closed. Post-emplacement monitoring must focus on using a combination of direct (e.g., pressure, temperature) and indirect methods (e.g., sonar surveys, simulation studies) as discussed in Section 3.1.3 to confirm containment of the injectate and any biogas produced to ensure durability. The requirements in this section should be followed until the closure of the cavern (see Section 3.5).

The Project Proponent/Operators must follow any post-emplacement requirements of the UIC or equivalents permit for the specified project, in addition to the following:

1. Monitoring of the cavern pressure (daily), temperature (6 monthly) and cavern volume (periodically). This can provide information regarding the cavern's natural closure characteristics and any ensuant pressure buildup, as well as information on mechanical integrity of the wellbore, salt creep-closure rate analysis, and flow rate calculation.
2. Intermittent gas monitoring for emissions of CO_2, CH_4, N, O_2 and VOCs at the wellhead (6 monthly) is required, where applicable and when gas volumes allow, until closure of the well. If any emissions outside of the normal average baseline range for any of these gases occur, then

further monitoring of gases should take place. Further monitoring entails measurements of:

a. Concentrations (e.g., CO_2, CH_4, N, O_2 and VOCs), usually measured by GC-MS
b. Stable isotope compositions of CO_2, CH_4 ($\delta^{13}C$ of CO_2, CH_4 and δD of CH_4), usually measured by an isotope ratio mass spectrometer (IRMS), to determine the source of the bio-gas emissions. $\delta^{13}C$ must be measured against the standard reference material Vienna Pee Dee Belemnite (VPDB). δD must be measured against the standard reference material Vienna Standard Mean Ocean Water (VSMOW).

3. Periodic surface subsidence monitoring, for example using GPS, which will be compared to baseline data and trends or reference points. This will be used to monitor for indication of salt movement or cavern roof collapse.
4. System Integrity monitoring should continue on a cadence specified by the UIC permit or equivalent . If no ongoing system integrity monitoring is specified, then external mechanical integrity tests should continue at the cadence specified during the operational phase, this can be changed as a result of information gathered during the operational/post-emplacement phase and with agreement from Isometric.

3.3
Risk of Reversal

Based on the present understanding, projects applicable to this Protocol are categorized as having a Very Low Risk Level of Reversal according to the Isometric Standard Risk Assessment Questionnaire. This is because salt caverns are impermeable (and thus act like a closed system). There should be no reversals unless there is a loss of cavern or well integrity,

and this technology does not yet have a documented history of reversals. A 2% buffer pool will be set aside as a precaution against CO_2 or CH_4 gas release or C release during brine displacement. This reversal risk will be reassessed when new scientific research and understanding arises.

Reversals will be accounted for by projects and the Isometric Registry as detailed in Section 5.6 of the Isometric Standard.

3.3.1 Attribution of reversals

When a reversal is detected and quantified, there are multiple considerations that will be taken into account to attribute the reversal to whatever has been emplaced in the cavern.

1. If the Project Proponent was the only entity emplacing into a given cavern, the Project Proponent will take on 100% of the reversal.
2. If the Project Proponent was one of multiple entities emplacing into that cavern, the Project Proponent will be allocated a percentage of the reversed CO_2 proportional to the mass of emplaced material. For example:
    - A cavern has a total of 200t of material emplaced at the time when the reversal is detected.
    - The Project Proponent has emplaced 50t of material in that salt cavern.
    - The amount of reversed CO_2 has been quantified to be 10t.
    - The Project Proponent must compensate for 25% (50/200) of 10t CO_2 = 2.5t of CO_2.

In instances where reversals are determined to be a result of negligence by the Operator or Project Proponent, project

crediting may be ceased.

3.4

Calculation of $CO_2e_{Emissions}$

$CO_2e_{Emissions}$ is the total greenhouse gas emissions associated with a given Reporting Period, RP, or batch, n.

Equations and emissions calculation requirements for $CO_2e_{Emissions}$, including considerations for monitoring activities, are set out in the relevant Protocol and are not repeated in this Module.

3.5

Closure Requirements

The Operator must ensure that all permit requirements associated with planning for, preceding with and monitoring of well or storage cavern closure are adhered to and documented as required by the permit. A Cavern Closure Plan must be prepared in accordance with the permit requirements. Site closure must follow this plan and any relevant regulatory jurisdiction requirements for site decommissioning. This must include plugging of any wells within the AOR.

CO_2 storage agreements with pore space owners will ensure activity in the storage cavern is prohibited for perpetuity following emplacement, ensuring that the injectate will not be subject to pressure disturbances (i.e, emplacement or production activities) in the salt cavern.

The Operator must monitor the cavern following emplacement completion to determine the three-dimensional extent of the injectate and cavern integrity and demonstrate that no injectate migration out of the salt cavern is occurring, as per

the post-emplacement monitoring plan in Section 3.2, above. In addition, due to the emplacement of biomass and bio-oil being a nascent field, if they are emplaced into a cavern alongside another waste material, the long-term effect of the multiple waste types, their interaction and impact on abandonment, has not been investigated and will require stringent monitoring both during emplacement, post emplacement and closure. Post-closure monitoring, for example, surface subsidence monitoring, may be required by the permit and all requirements must be followed.

Prior to closure, an assessment must be completed to demonstrate that cavern integrity will be maintained (for example proving the absences of salt fracture generation) and the emplaced material can be considered stabilized, eliminating the risk of migration or release of the injectate or its degradation products from the cavern to the atmosphere. The cavern integrity assessment must be conducted in the following ways:

1. Measurement of Cavern pressure to ensure it is not pressurized. It must be demonstrated that any natural increase in pressure in the closed cavity due to long-term convergence does not exceed the fracture pressure.
2. Alternatively emerging technologies for measuring cavern integrity can be used, and will be outlined in subsequent Protocol versions and as measurement and monitoring technologies advance.

If the cavern integrity can be demonstrated by the above methods, and is independently reviewed and certified by a registered Professional Geologist (i.e., Chartered Geologist or equivalent), the emplaced material and salt cavern will be

considered stabilized and additional monitoring post-closure may be discontinued if allowed under the applicable permit.

The long term stability of the cavern system should also be monitored by measuring surface movement over a period determined and agreed with the UIC or equivalents governing body and will be subject to change. The UIC or equivalents executive director or equivalent may extend the period of post closure monitoring if they determine that the well or cavern may endanger an underground source of drinking water or freshwater aquifer and subject to ongoing analysis.

3.5.1
Record keeping

All records associated with the characterization, design, construction, injection operation, monitoring, and cavern closure must be developed, reported in the project design document, to the VVB's and to proper authorities as required by the permit.

All records must be maintained for a minimum of 10 years after well closure. All closure and post-closure monitoring records must be maintained by the Project Proponent for a minimum of 10 years after closure.

4.0

Definitions and Acronyms

Bio-oil
A mixture of water, organic acids, aldehydes, ketones, sugars, phenols, and other organic compounds derived from the thermal breakdown of biomass. Thermal breakdown of biomass is achieved via thermochemical processes, such as pyrolysis, which heat biomass in low- or no-oxygen environments to high temperatures (~e.g. 350-650°C). Bio-oil is often also referred to as pyrolysis oil or bio-crude.

Buffer Pool
A common and recognized insurance mechanism among Registries allowing Credits to be set aside (in this case by Isometric) to compensate for Reversals which may occur in the future.

Crediting Period
The period of time over which a Project Design Document is valid, and over which Removals or Reductions may be Verified, resulting in Issued Credits.

Durability
The amount of time carbon removed from the atmosphere by an intervention – for example, a CDR project – is expected to reside in a given Reservoir, taking into account both physical risks and socioeconomic constructs (such as contracts) to protect the Reservoir in question.

Feedstock
Raw material which is used for CO_2 Removal or GHG Reduction.

International Standards Organization (ISO)
A worldwide federation (NGO) of national standards bodies from more than 160 countries, one from each member country.

Module

Independent components of Isometric Certified Protocols which are transferable between and applicable to different Protocols.

Project
An activity or process or group of activities or processes that alter the condition of a Baseline and leads to Removals or Reductions.

Project Design Document (PDD)
The document that clearly outlines how a Project will generate rigorously quantifiable Additional high-quality Removals or Reductions.

Project Proponent
The organization that develops and/or has overall legal ownership or control of a Removal or Reduction Project.

Protocol
A document that describes how to quantitatively assess the net amount of CO_2 removed by a process. To Isometric, a Protocol is specific to a Project Proponent's process and comprised of Modules representing the Carbon Fluxes involved in the CDR process. A Protocol measures the full carbon impact of a process against the Baseline of it not occurring.

Registry
A database that holds information on Verified Removals and Reductions based on Protocols. Registries Issue Credits, and track their ownership and Retirement.

Reservoir
A location where carbon is stored. This can be via physical barriers (such as geological formations) or through partitioning based on chemical or biological processes (such as mineralization or photosynthesis).

Reversal
The escape of CO_2 to the atmosphere after it has been stored, and after a Credit has been Issued. A Reversal is classified as avoidable if a Project Proponent has influence or control over it and it likely could have been averted

through application of reasonable risk mitigation measures. Any other Reversals will be classified as unavoidable.

Standards (scientific)

Standard physical constants as well as standard values set forth by bodies such as the National Institute of Standards and Technology (NIST) or others.

Storage

Describes the addition of carbon dioxide removed from the atmosphere to a reservoir, which serves as its ultimate destination. This is also referred to as “sequestration”.

Validation and Verification Bodies (VVBs)

Third-party auditing organizations that are experts in their sector and used to determine if a project conforms to the rules, regulations, and standards set out by a governing body. A VVB must be approved by Isometric prior to conducting validation and verification.

5.0

Appendix 1: Monitoring Plan Requirements

This appendix details how the Project Proponent must monitor, document and report all metrics identified within this Module to demonstrate the durability of CO_2 removal. Following this guidance will ensure the Project Proponent measures and confirms CO_2 removed and long-term storage compliance, and will enable quantification of the emissions removal resulting from the Project activity during the Project Crediting Period, prior to each Verification.

This methodology utilizes a comprehensive monitoring and documentation framework that captures the GHG impact in each stage of a Project. Monitoring and detailed accounting practices must be conducted throughout to ensure the continuous integrity of the net CO_2e and crediting.

The Project Proponent must develop and apply a monitoring plan according to ISO 14064-2 principles of transparency and accuracy that allows the quantification and proof of GHG emissions removals.

PARAMETER	PARAMETER DESCRIPTION	MONITORING PHASE	REQUIRED
pH	pH of displaced brine on the first day of injection to represent background cavern conditions	Pre Injection	Always
Temperature	Temperature of displaced brine on the first day of injection to represent background cavern conditions	Pre Injection	Always

PARAMETER	PARAMETER DESCRIPTION	MONITORING PHASE	REQUIRED
Conductivity or other salinity measurement	Conductivity of displaced brine on the first day of injection to represent background cavern conditions	Pre Injection	Always
TOC of the formation water	TOC of displaced brine on the first day of injection to represent background cavern conditions	Pre Injection	Always
pH	pH of injection slurry	Operation	Always
Temperature	Temperature of injection slurry	Operation	Always
Conductivity or other salinity measurement	Conductivity/Chloride content or other alternative determination as required by the regulator of the injection slurry	Operation	Always
δ^{13} Carbon signature	δ^{13} Carbon signature of the compounds of the biomass	Operation	Not required but helpful
Internal Well Integrity (Corrosion monitoring)	Demonstration of internal mechanical integrity	Operation & Post Injection	Always

PARAMETER	PARAMETER DESCRIPTION	MONITORING PHASE	REQUIRED
External Well integrity	Demonstration of external mechanical integrity	Operation & Post Injection	Always
Temperature	Well bottom (Cavern) Temperature	Post Injection	Always
Pressure	Well bottom (Cavern) Pressure	Operation & Post Injection	Always
Sonar Survey	Sonar surveys to determine cavern volume and fill	Operation & Post Injection	Always
Cavern Fill	Determination of fill depth and its evolution	Pre Injection	Always
Total carbon content	%wt of C in the displaced brine	Operation	Always
Temperature	Temperature of the displaced brine	Operation	Always

PARAMETER	PARAMETER DESCRIPTION	MONITORING PHASE	REQUIRED
pH	pH of the displaced brine	Operation	Always
Conductivity or other salinity measurement	Chloride content/conductivity of displaced brine	Operation	Always
Biomass constituents	Biomass constituents e.g., water, oxygen and nitrogen	Operation	Not required but helpful
Gas composition	Gas composition analysis CO_2, N, O_2, CH_4, or VOC levels in well headspace with resolution of at least 0.01%vol	Operation & Post Injection	Under certain conditions: • When gas is detected and mea not a result of any initial gas re pre-existing gas stores within t
Surface Subsidence	Surface Subsidence	Operation & Post Injection	Always
Surface topography or determination of reference point	Surface topography or reference set point as a baseline for subsidence monitoring	Pre Injection	Always

PARAMETER	PARAMETER DESCRIPTION	MONITORING PHASE	REQUIRED
USDW quality	Monitoring of USDWs for increases in pressure and changes to chloride concentrations at a minimum	Operation & Post Injection	Required as per permit require helpful
Gas isotope stability	Stable isotope compositions of carbon containing species - usually measure by IRMS	Operation & Post Injection	Under certain conditions: • When gas is detected and mea not a result of any initial gas re pre-existing gas stores within t
Cavern volume	Cavern volume prior to injection	Pre Injection	Always
Permeability	Permeability of the salt	Pre Injection	Always
Porosity & permeability	Porosity & Permeability of caprock and any interbeds	Pre Injection	Always
Subsurface features	Investigate pre-existing faults and presence of any interbeds	Pre Injection	Always

6.0

Relevant Works

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7. Davidson, B.C., Dusseault, M.B., Lemieux, B., 1997. Design And Management Of Salt Solution Caverns For Toxic Waste Disposal. https://doi.org/10.2118/97-151 ↩
8. Area of Review (AOR) is the representation of the cavern extent on the ground surface; a surface delineation of subsurface pressure influence resulting from emplaced materials ↩ ↩
9. API Standard 65- part 2 ↩ ↩ ↩ [2]
10. Influence Area is the area defined for subsidence monitoring by the regulatory agency or operator, if not otherwise specified in the permit ↩
11. For Class V wells, the well must be permitted and not 'authorized by rule', and must consider the specific

emplacement and durable storage of bio-oil and/or biomass in the geologic reservoir. As of writing, the utilization of Class V wells should be limited to wells operating under the Other / Experimental category of Class V wells or other appropriate well type as approved by the UIC permitting authority. ↩ ↩️

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Contributors

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Changelog

v1.2

PENDING CERTIFICATION

v1.1

CERTIFIED

Key requirement changes include updates to the Well Construction section to ensure the integrity of pre-existing wells and the addition of seismic monitoring in the Migration and Storage Reversal Monitoring section.

v1.0

CERTIFIED

Initial Release.

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